

Multi-Agent Pipeline Report

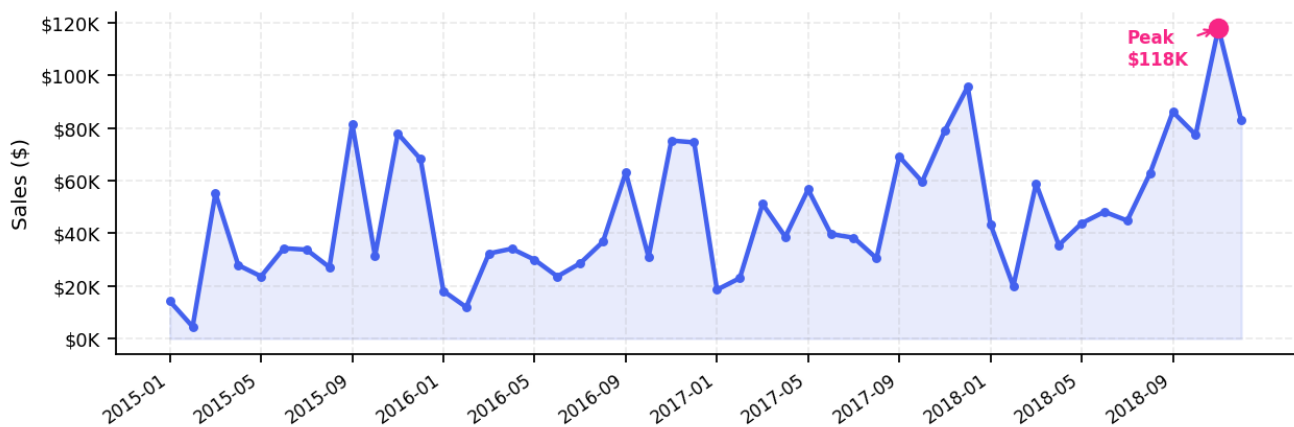
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Winning model: SARIMAX · Critic score: 8/10

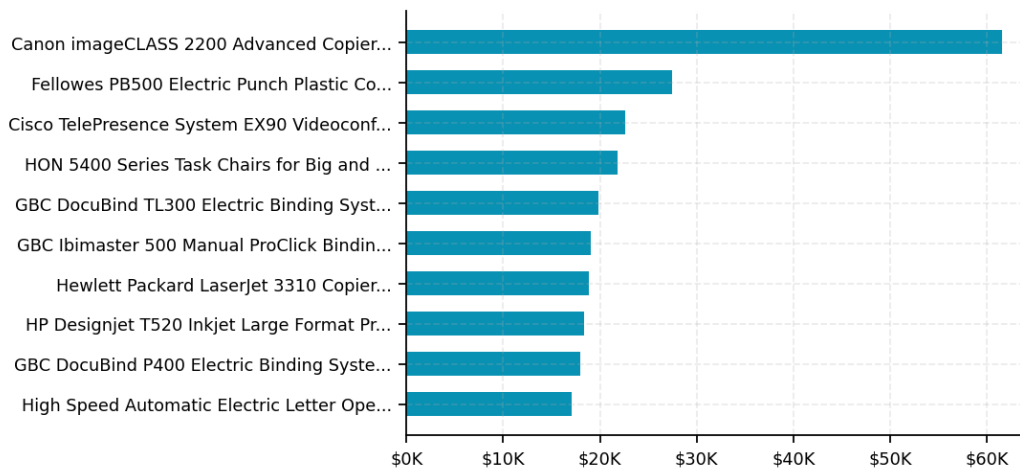
Dataset Overview

Metric	Value
Total order lines	9,800
Date range	2015-01-03 → 2018-12-30
Sales mean / median	\$230.77 / \$54.49
Outliers / Missing	123 / None

Monthly Sales Trend (2015-2018)



Top 10 Products by Revenue



Pipeline Run Summary

Status	Notes
success	The data quality report indicates that out of 9,800 total rows, there are no missing values or duplicate rows, but there
success	EDA complete. 6 charts. 3 anomalous weeks. Holiday lift: 4.7%
success	Prophet MAE=19145 MAPE=24.16% SARIMAX MAE=17892 MAPE=23.06%
success	Winner: SARIMAX (Prophet MAPE=24.16% SARIMAX MAPE=23.06%)
success	Narrative generated (463 words).
approved	Score=8/10

Data Quality Gate

Metric	Value
Total rows	9800
Date range	2015-01-03 → 2018-12-30
Missing values	{}
Duplicates removed	0
Outliers flagged ($z > 3\sigma$)	123

Executive Summary

\$861K

2019 total forecast

\$118K

Peak month forecast

12

Months forecast

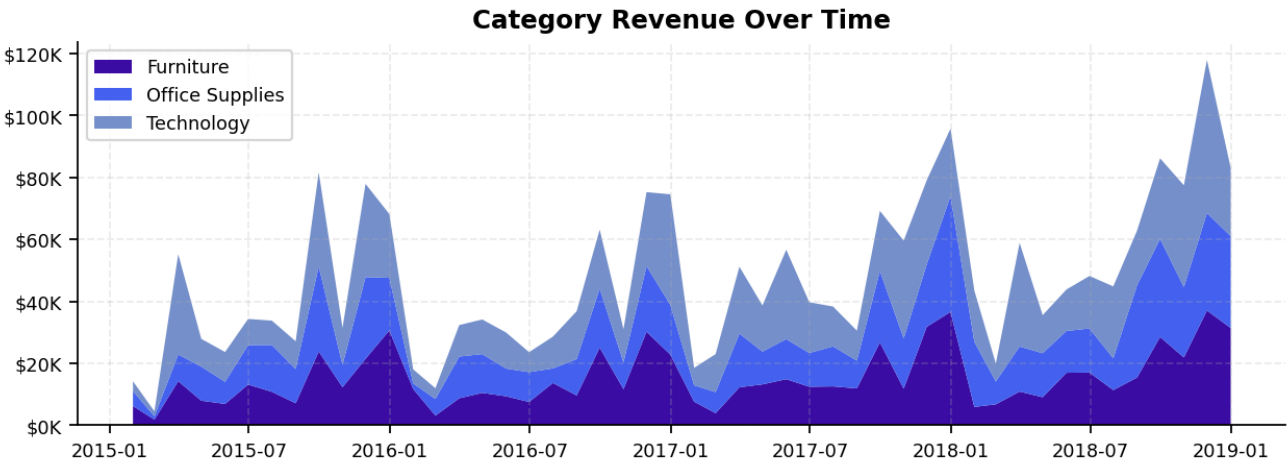
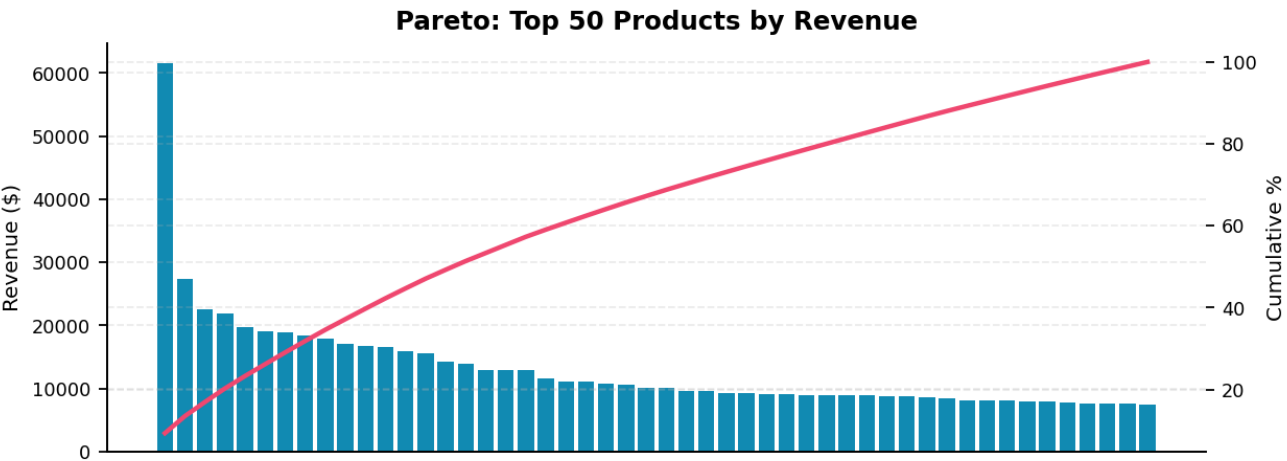
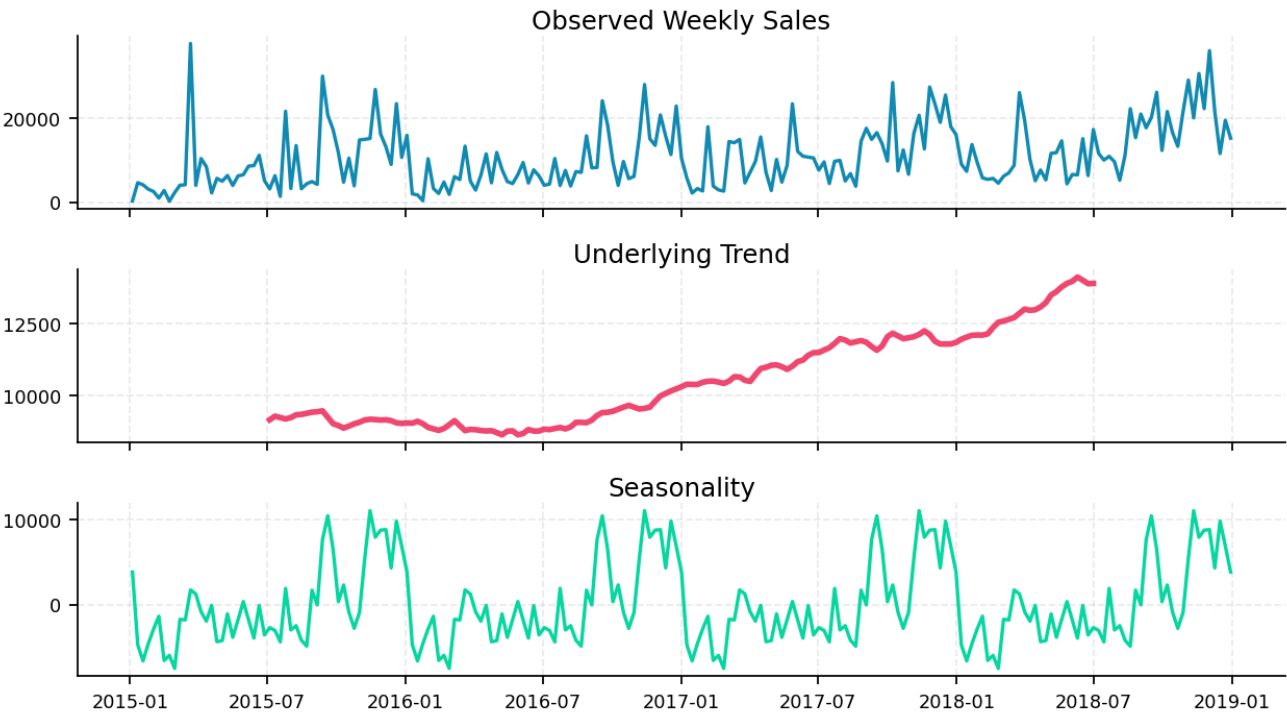
SARIMAX

Winning model

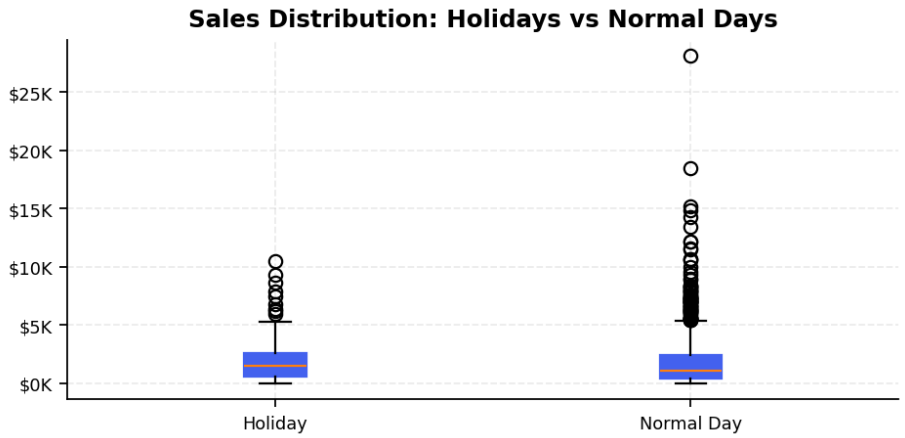
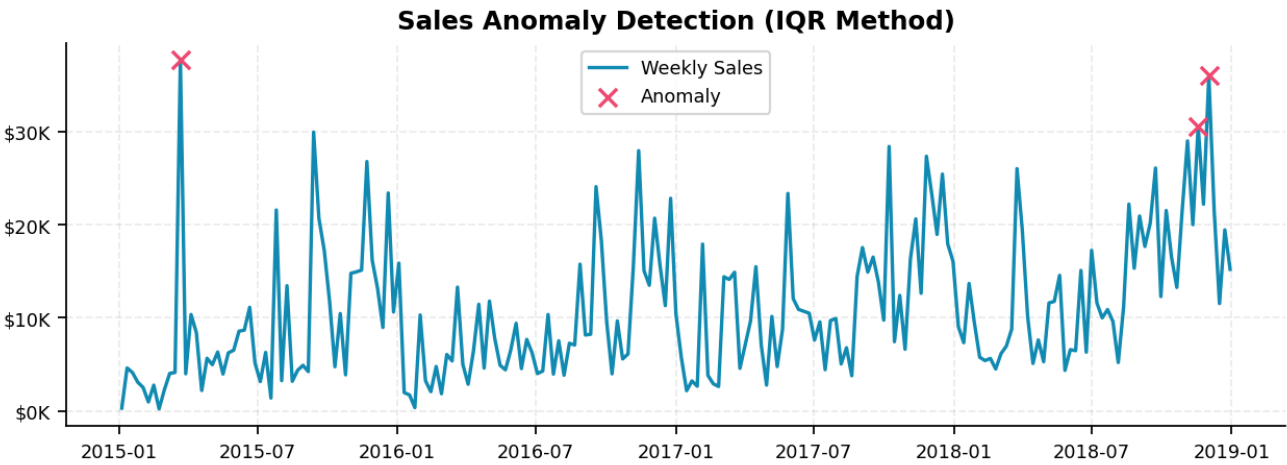
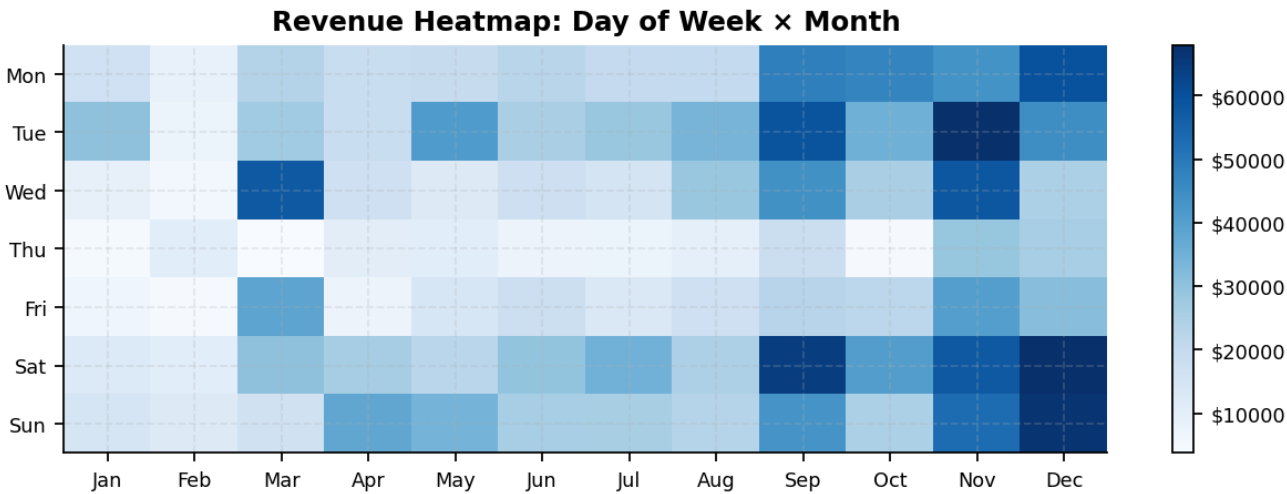
8/10

Critic score

Exploratory Data Analysis (EDA)



Seasonality & Anomalies



12-Month Forecast — SARIMAX

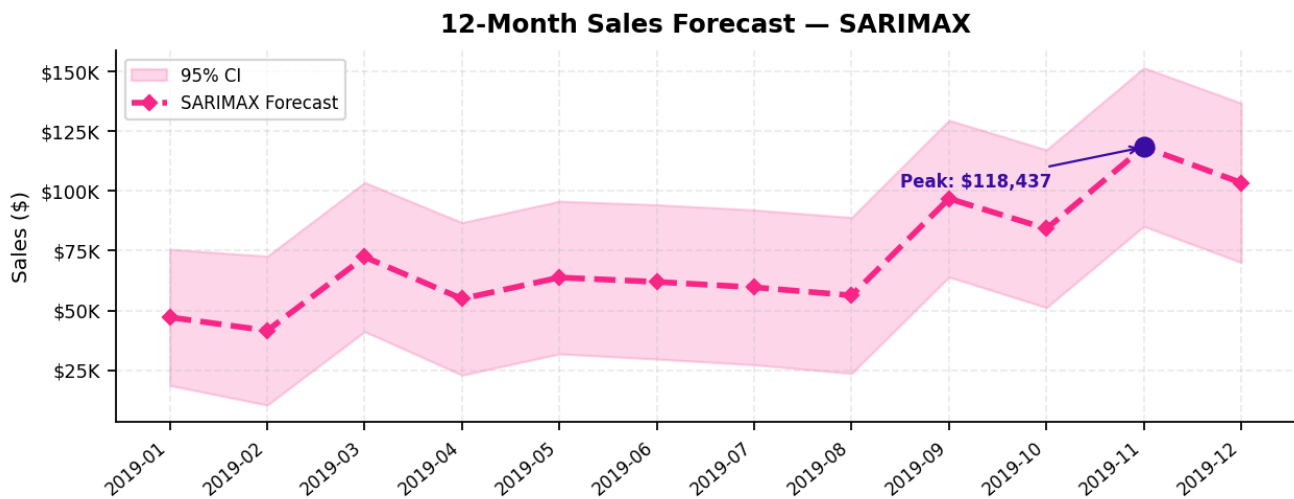


Figure 4 — SARIMAX model forecast for Jan–Dec 2019 with 95% confidence interval.

DS	YHAT	YHAT LOWER	YHAT UPPER	MODEL
2019-01	\$47,104	\$18,567	\$75,641	SARIMAX
2019-02	\$41,564	\$10,372	\$72,756	SARIMAX
2019-03	\$72,388	\$41,138	\$103,638	SARIMAX
2019-04	\$54,862	\$22,900	\$86,825	SARIMAX
2019-05	\$63,801	\$31,800	\$95,802	SARIMAX
2019-06	\$61,963	\$29,629	\$94,296	SARIMAX
2019-07	\$59,675	\$27,220	\$92,130	SARIMAX
2019-08	\$56,296	\$23,609	\$88,984	SARIMAX
2019-09	\$96,847	\$63,999	\$129,695	SARIMAX
2019-10	\$84,248	\$51,200	\$117,296	SARIMAX
2019-11	\$118,437	\$85,215	\$151,660	SARIMAX
2019-12	\$103,483	\$70,072	\$136,893	SARIMAX
TOTAL	\$860,669			

Forecast Model Comparison

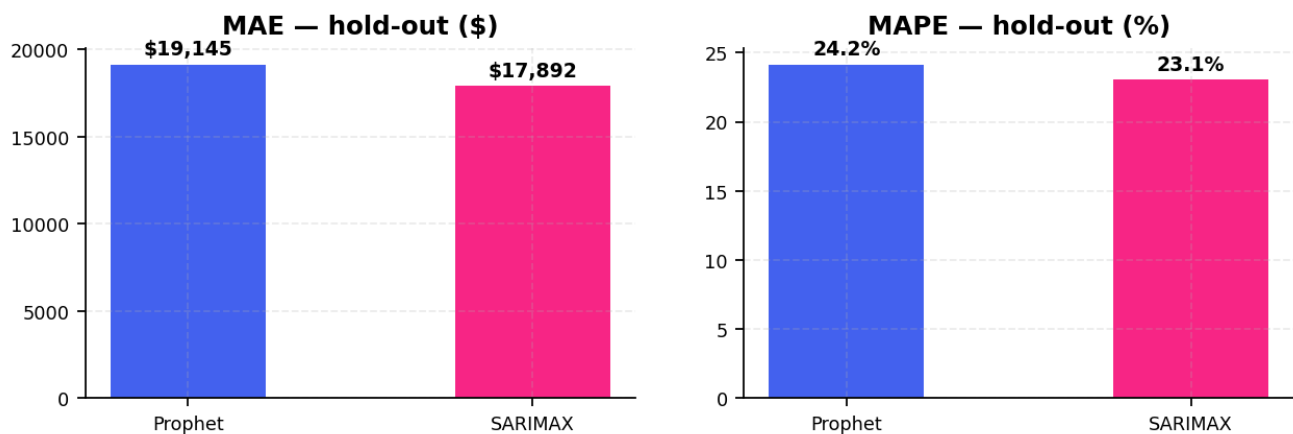


Figure 5 — MAE and MAPE on hold-out evaluation set.

Winner: SARIMAX

The winning model, SARIMAX, outperformed the Prophet model on this dataset with a Mean Absolute Percentage Error (MAPE) of 23.0561 compared to Prophet's 24.1612, indicating a more accurate forecast for our specific series characterized by strong seasonality and an upward trend. One trade-off of using SARIMAX is its complexity; while it achieved a lower Mean Absolute Error (MAE) of 17,891.72, it requires careful tuning of parameters and may not generalize as well to datasets with different characteristics compared to the more flexible Prophet model.

Model	MAE (\$)	RMSE (\$)	MAPE (%)	AIC
Prophet	19144.69	21093.05	24.1612	N/A
SARIMAX	17891.72	22195.1	23.0561	449.42

Business recommendation: Use SARIMAX for the 12-month planning forecast. Re-evaluate both models in 6 months with actuals to track accuracy drift.

Strategic Recommendations

1

Enhance Marketing for Technology Products

: Increase targeted marketing efforts for Technology, which generated **\$827,455.87** in sales, by 20% by Q2 2019.

2

Optimize Inventory for Peak Months

: Ensure inventory levels are sufficient for November and December, projected at **\$118,437.42** and **\$103,482.94** respectively, by Q3 2019.

3

Investigate Anomalous Sales Periods

: Conduct a detailed analysis of the sales anomalies from March 2015, November 2018, and December 2018 by Q1 2019 to identify root causes.

4

Leverage Holiday Lift

: Develop promotional campaigns that capitalize on the **4.74%** holiday lift, launching by October 2019 to maximize sales during the holiday season.

5

Monitor Seasonal Trends

: Regularly review the seasonality index, particularly for low-performing months like February, to adjust marketing strategies accordingly by Q1 2019. This report provides a comprehensive overview of past performance and actionable insights for future growth.